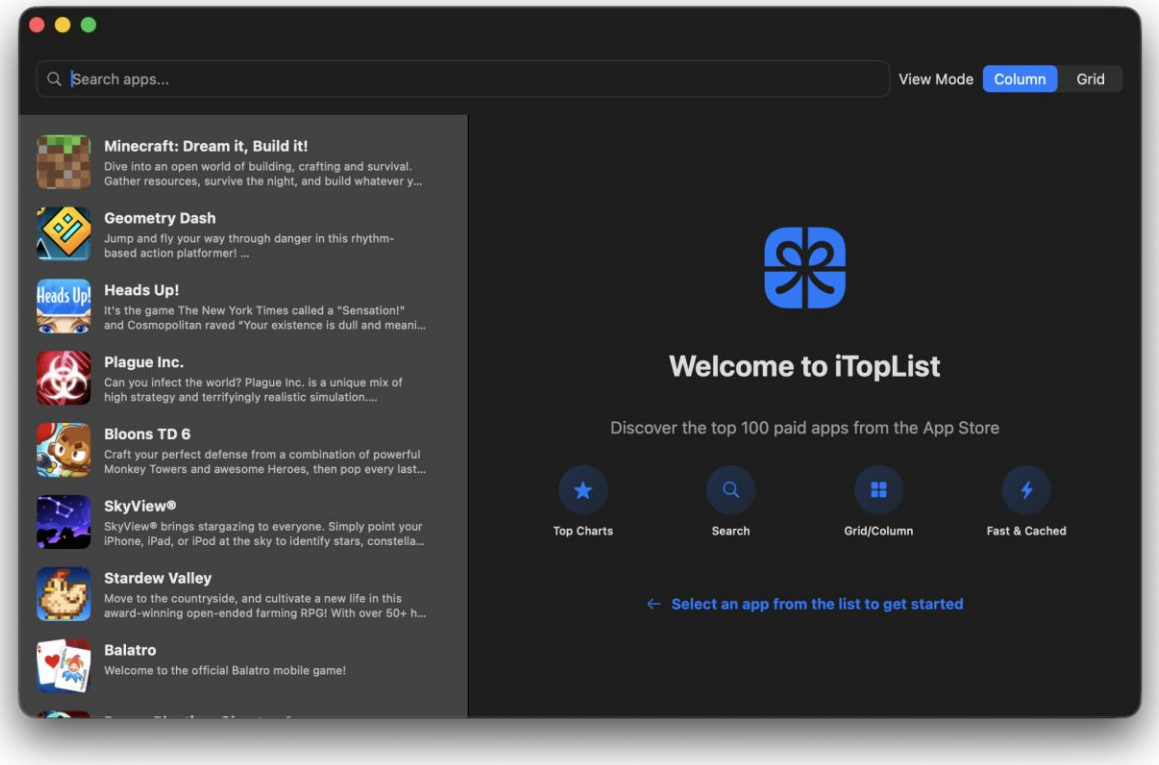
# iTopList - Top Paid Apps Browser for macOS

## Project Overview

iTopList is a native macOS application built with SwiftUI that displays the top 100 paid applications from the Apple App Store. The app fetches data from Apple's RSS feed, provides a clean dual-pane interface for browsing apps, and implements comprehensive caching for offline access.

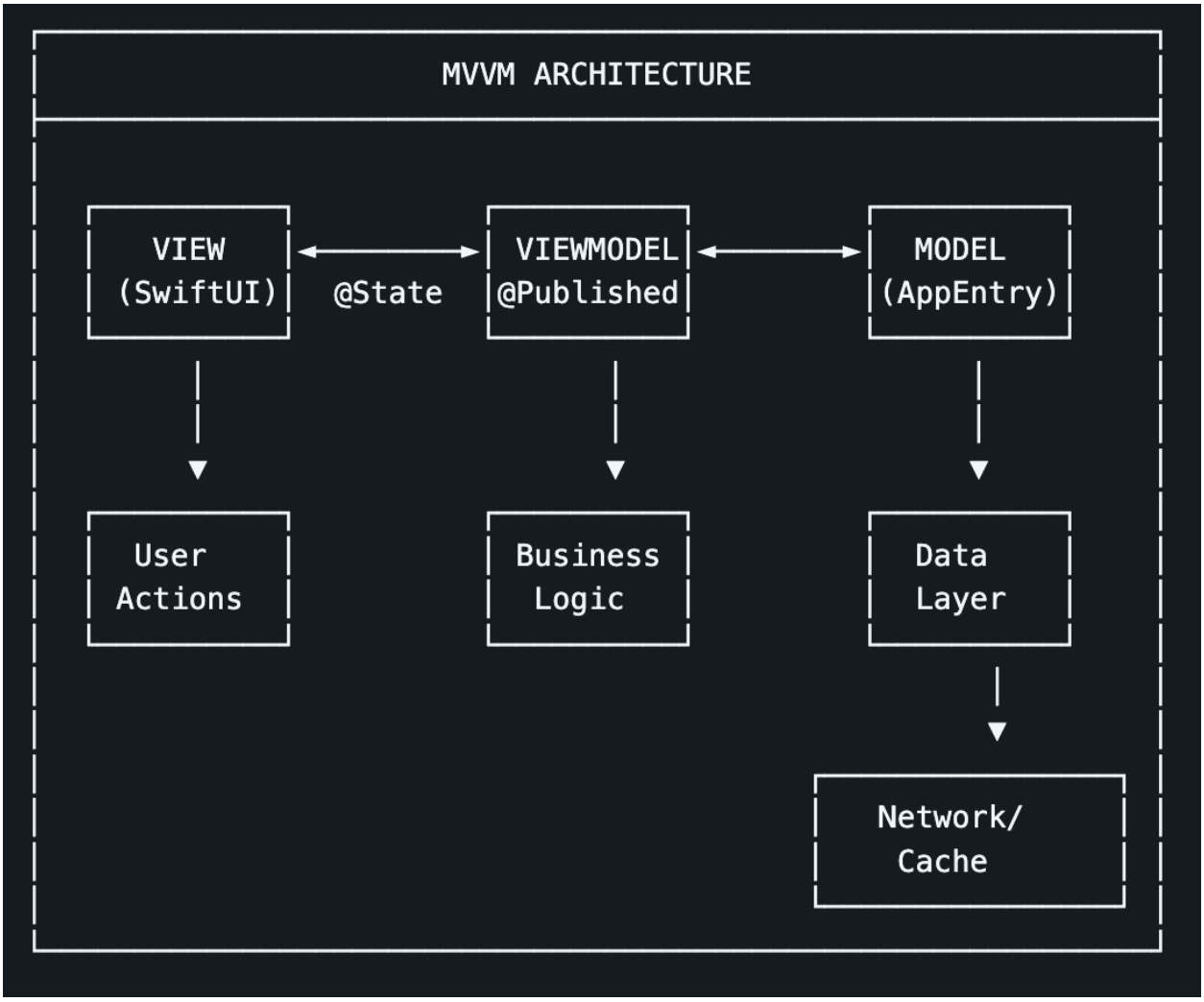



## Features

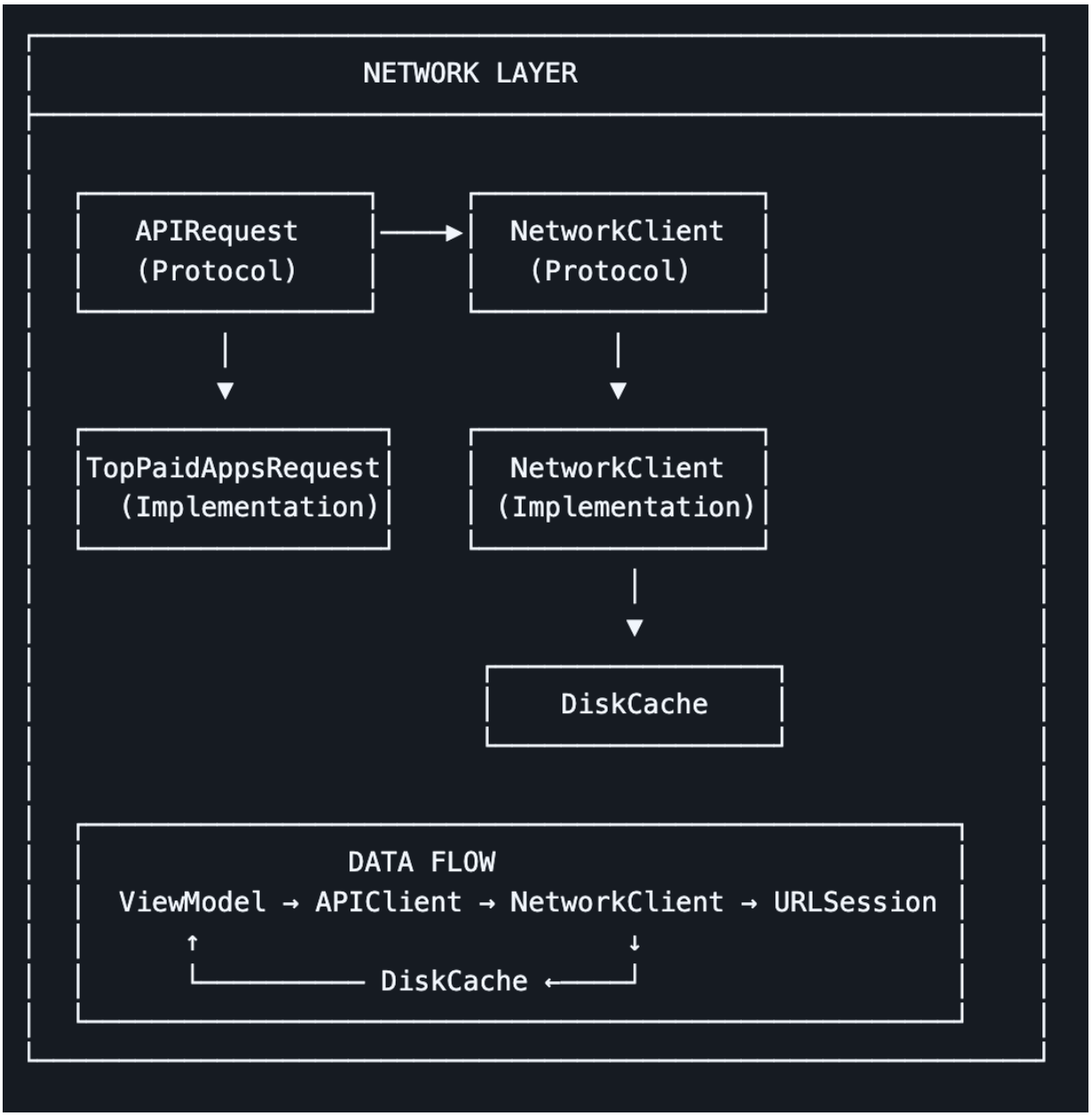
| Feature | Description |
|---|---|
| App Browsing | Browse top 100 paid apps from Apple's RSS feed |
| Dual-Pane Interface | Master-detail layout similar to macOS Settings |
| View Modes | Toggle between list and grid layouts |
| Live Search | Real-time filtering with debounced search |
| Infinite Scroll | Pagination with 20 items per page |
| Offline Support | Full offline access with disk caching |
| Image Caching | Two-tier cache (memory + disk) for icons |

## Architecture

### MVVM Pattern



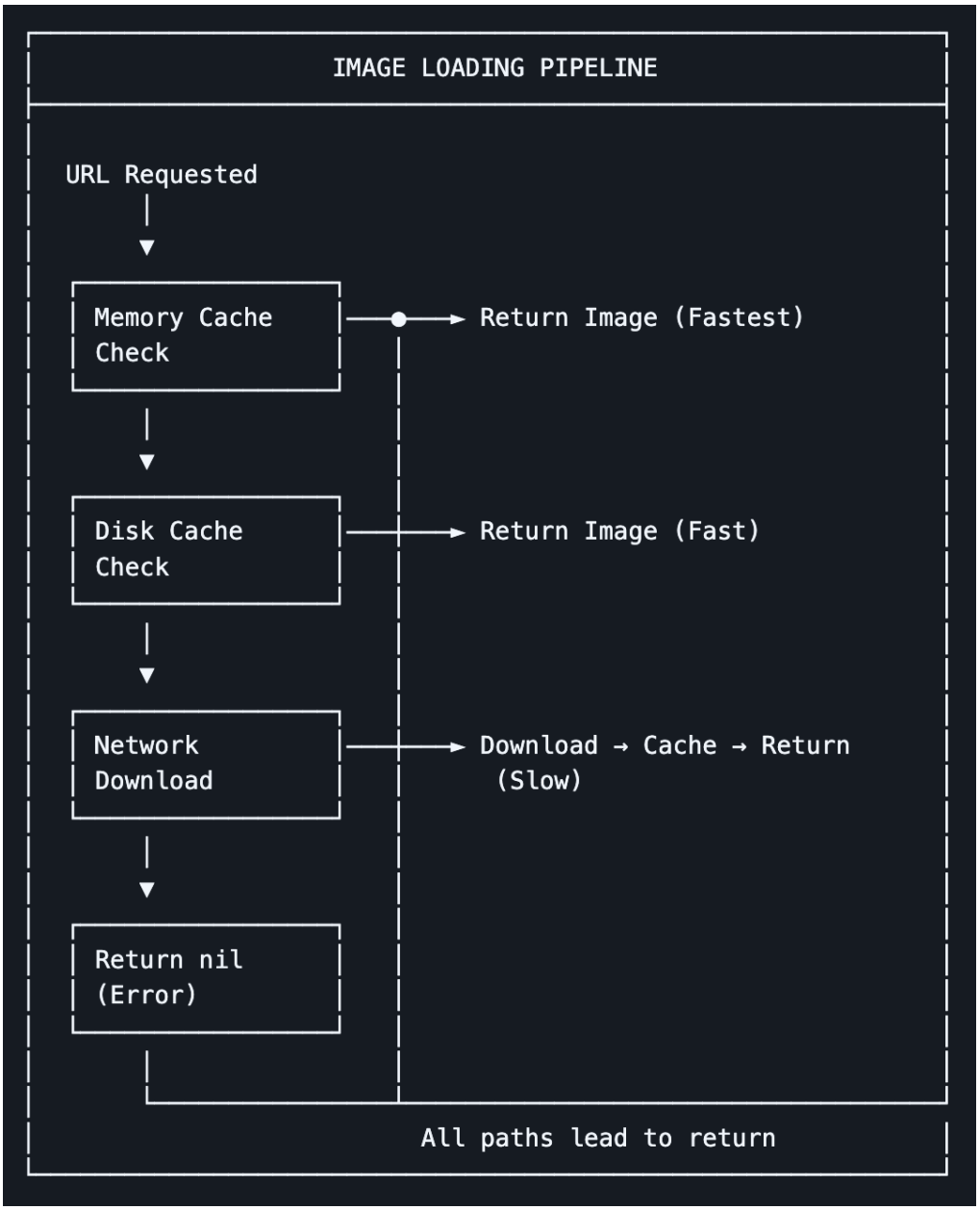
**Network Layer Architecture**



## Project Structure

```
iTopList/
├── App/
│   └── iTopListApp.swift                # App entry point with window configuration
│
├── Models/
│   ├── RSSResponse.swift                 # Root RSS feed response
│   ├── AppEntry.swift                     # Main app model with Hashable conformance
│   ├── Label.swift                        # Generic label wrapper for strings
│   └── ImageData.swift                     # Image URL container
│
├── Networking/
│   ├── API/
│   │   ├── HTTPMethod.swift                # GET, POST, PUT, DELETE enum
│   │   ├── APIRequest.swift                 # Generic request protocol with associated type
│   │   └── TopPaidAppsRequest.swift         # Concrete request for top paid apps
│   │
│   ├── Client/
│   │   ├── NetworkError.swift               # Custom error types with descriptions
│   │   ├── NetworkClientProtocol.swift       # Protocol for dependency injection
│   │   ├── NetworkClient.swift               # URLSession implementation with caching
│   │   ├── APIClientProtocol.swift            # Business layer protocol
│   │   └── APIClient.swift                     # High-level API client
│   │
│   └── Cache/
│       └── DiskCache.swift                     # File system cache with thread safety
│
├── Services/
│   └── ImageLoader.swift                        # Async image loader with two-tier cache
│
├── ViewModels/
│   └── AppListViewModel.swift                    # Main view model with pagination logic
│
├── Views/
│   ├── ContentView.swift                         # Root view with state management
│   ├── ColumnView.swift                          # HSplitView list layout
│   ├── GalleryView.swift                         # Grid layout with LazyVGrid
│   ├── WelcomeView.swift                          # Empty state with features grid
│   ├── AppRow.swift                               # List item with icon and text
│   ├── GalleryItemView.swift                      # Grid item with icon
│   ├── AppIconView.swift                          # Reusable async image loader
│   ├── DetailView.swift                           # App details with close button
│   └── Components/
│       └── FeatureItem.swift                      # Feature grid item for welcome screen
│
└── Tests/
    ├── Mocks/
    │   ├── MockNetworkClient.swift                 # Network mock for testing
    │   ├── MockAPIClient.swift                      # API mock for testing
    │   └── MockDiskCache.swift                       # In-memory cache mock
    ├── TestHelpers/
    │   └── TestData.swift                            # Test data factory methods
    ├── NetworkingTests/
    │   ├── NetworkLayerTests.swift                   # Request/response tests
    │   ├── APIClientTests.swift                      # API client tests
    │   └── DiskCacheTests.swift                      # Cache operation tests
    ├── ViewModelTests/
    │   └── AppListViewModelTests.swift               # State and pagination tests
    └── ServiceTests/
        └── ImageLoaderTests.swift                    # Image loading tests
```

### Image Loading Pipeline




### Performance Optimizations

1. Image Caching

- Memory Cache: 100 image limit, 50MB cost limit
- Disk Cache: Persistent storage with key sanitization
- Preloading: Background preload for upcoming items

2. Search Debouncing

3. Memory Management

- Clears cache on memory warnings
- Task cancellation for disappearing cells
- Weak references in closures to prevent retain cycles

4. Lazy Loading

- Images load only when cells appear
- Pagination loads only visible items
- LazyVGrid for efficient grid rendering
-

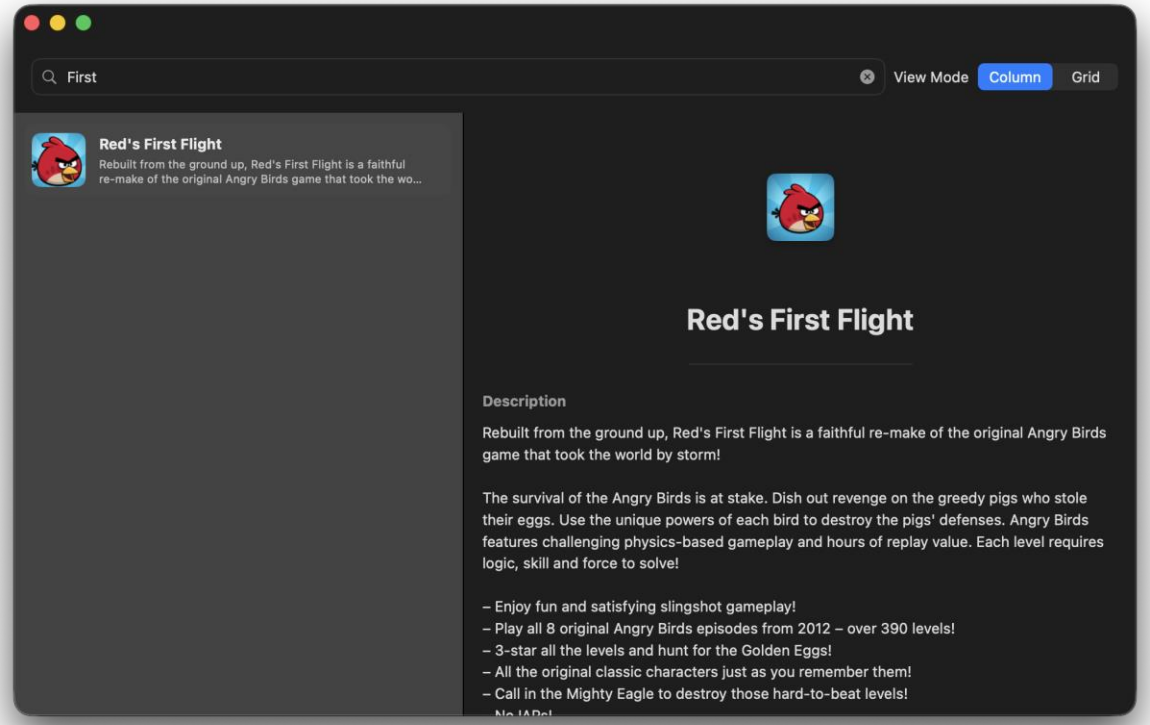
Build Requirements

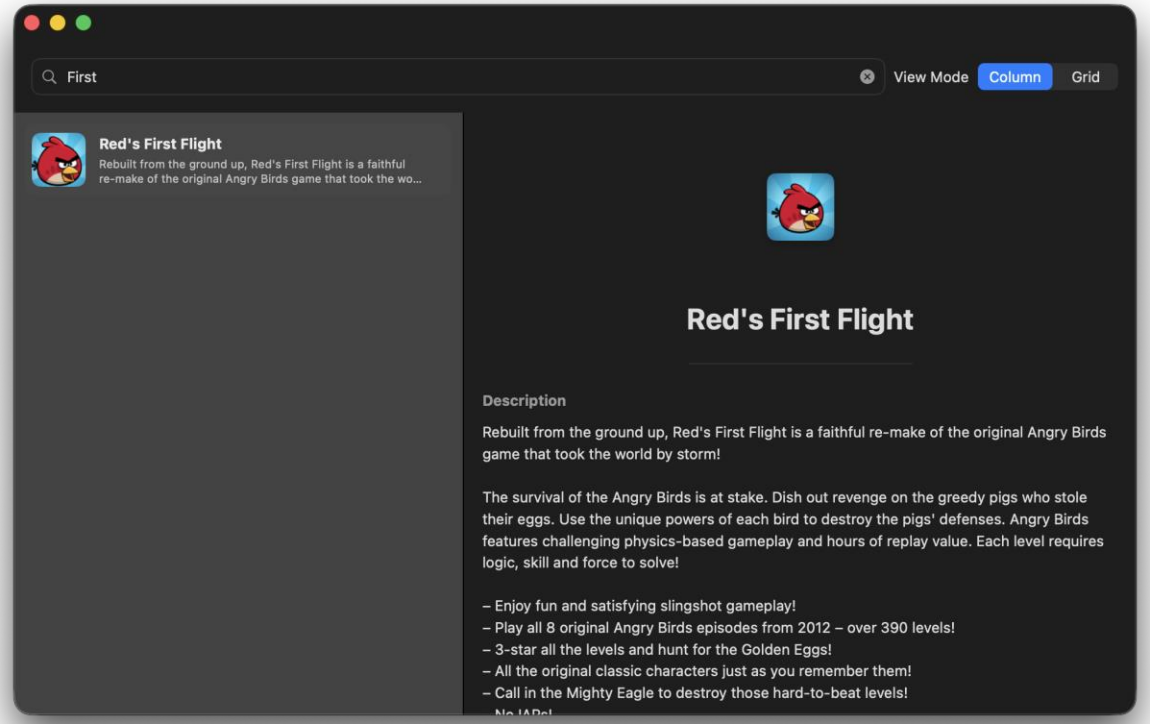
- Xcode Version: 14.0 or later
- Swift Version: 5.7 or later

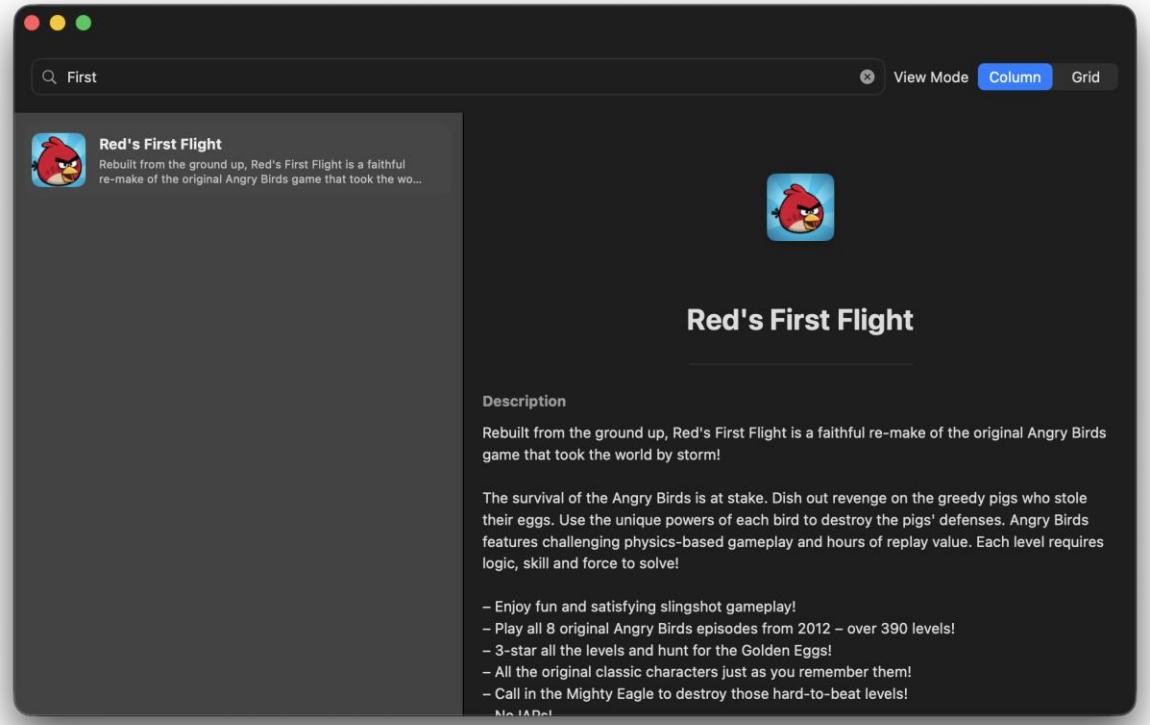
Dependencies

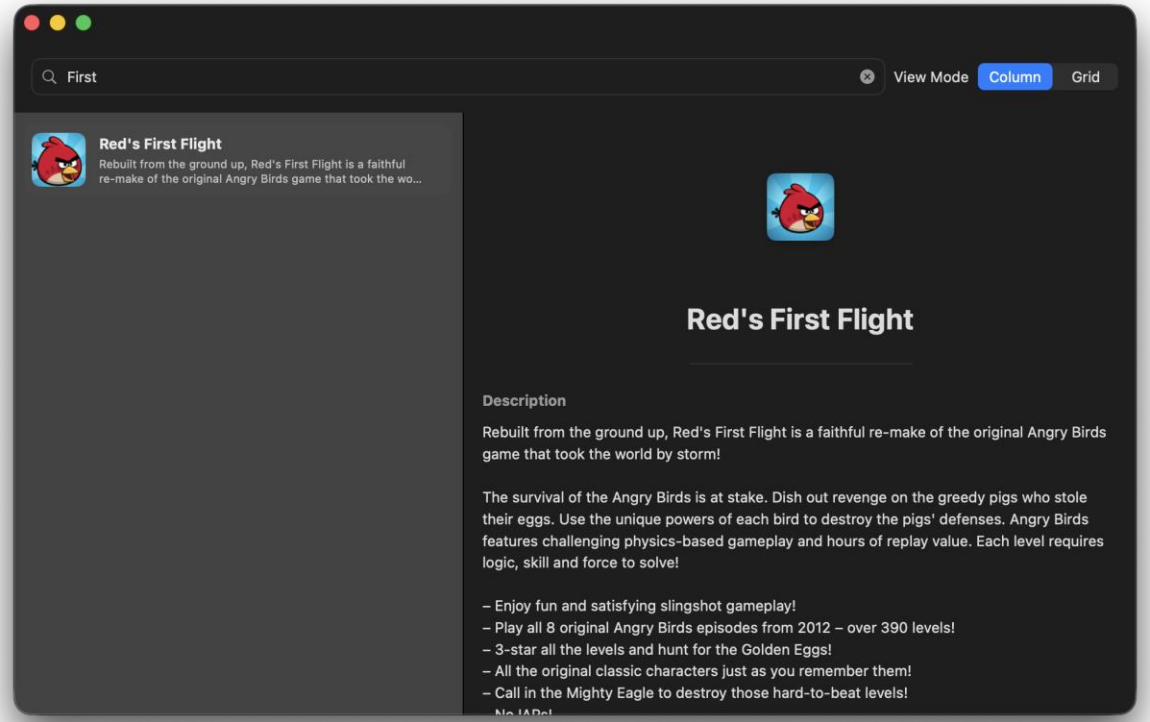
- No third-party dependencies (100% Apple SDKs)
- Uses SwiftUI, AppKit, Combine frameworks

## Screenshots:




First
View Mode
Column
Grid
Red's First Flight
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the wo...
Red's First Flight
Description
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the world by storm!
The survival of the Angry Birds is at stake. Dish out revenge on the greedy pigs who stole their eggs. Use the unique powers of each bird to destroy the pigs' defenses. Angry Birds features challenging physics-based gameplay and hours of replay value. Each level requires logic, skill and force to solve!
– Enjoy fun and satisfying slingshot gameplay!
– Play all 8 original Angry Birds episodes from 2012 – over 390 levels!
– 3-star all the levels and hunt for the Golden Eggs!
– All the original classic characters just as you remember them!
– Call in the Mighty Eagle to destroy those hard-to-beat levels!


First
View Mode
Column
Grid
Red's First Flight
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the wo...
Red's First Flight
Description
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the world by storm!
The survival of the Angry Birds is at stake. Dish out revenge on the greedy pigs who stole their eggs. Use the unique powers of each bird to destroy the pigs' defenses. Angry Birds features challenging physics-based gameplay and hours of replay value. Each level requires logic, skill and force to solve!
– Enjoy fun and satisfying slingshot gameplay!
– Play all 8 original Angry Birds episodes from 2012 – over 390 levels!
– 3-star all the levels and hunt for the Golden Eggs!
– All the original classic characters just as you remember them!
– Call in the Mighty Eagle to destroy those hard-to-beat levels!


First
View Mode
Column
Grid
Red's First Flight
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the wo...
Red's First Flight
Description
Rebuilt from the ground up, Red's First Flight is a faithful re-make of the original Angry Birds game that took the world by storm!
The survival of the Angry Birds is at stake. Dish out revenge on the greedy pigs who stole their eggs. Use the unique powers of each bird to destroy the pigs' defenses. Angry Birds features challenging physics-based gameplay and hours of replay value. Each level requires logic, skill and force to solve!
– Enjoy fun and satisfying slingshot gameplay!
– Play all 8 original Angry Birds episodes from 2012 – over 390 levels!
– 3-star all the levels and hunt for the Golden Eggs!
– All the original classic characters just as you remember them!
– Call in the Mighty Eagle to destroy those hard-to-beat levels!